

Fast Dense Solvers for Spectral Physics-Informed Neural Networks via Gauss–Newton Compression

A Sherman–Morrison–Woodbury and Matrix-Free LSQR Study
on One-Dimensional Boundary-Value Problems

Research Report

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Abstract

We study Gauss–Newton (GN) optimisation of spectral physics-informed neural networks (PINNs) for two boundary-value problems on $[-1, 1]$ with homogeneous Dirichlet conditions: the linear problem $-u'' = f$ and the nonlinear problem $-u'' + u^3 = f$. The solution is parameterised as $u_\theta(x) = \Phi(x) c(\theta)$ where Φ is a Chebyshev basis evaluated at Chebyshev–Gauss–Lobatto (CGL) collocation nodes and $c(\theta) \in \mathbb{R}^K$ is the output of a small multilayer perceptron with P parameters. Because $\text{rank}(\partial r / \partial \theta) \leq K \ll P$, the GN normal equations are rank-deficient and a Levenberg–Marquardt (LM) damping is required. We compare three solvers of the damped normal system: (i) a strong dense baseline using LAPACK `lstsq` (linear) or SciPy `least_squares` with the analytic Jacobian (nonlinear); (ii) a Sherman–Morrison–Woodbury (SMW) compression that reduces the $P \times P$ system to a $K \times K$ solve; and (iii) a matrix-free LSQR solver that never materialises the Jacobian. On the nonlinear problem at $N = 2000$ collocation points, the SMW method is $209\times$ **faster** than the dense baseline (786 ms vs 164 s) at comparable relative L^2 accuracy ($\sim 10^{-6}$ vs 10^{-11}); the matrix-free LSQR variant gives a $111\times$ speedup at the same N . Ablation studies sweeping the basis truncation K , the LSQR inner-iteration count, and the network initialisation reveal that zero-initialisation of the output layer is essential at large N on the nonlinear problem (Xavier-init fails outright at $N = 1000$), and that LSQR’s matrix-free advantage does *not* transfer to large K on this problem class because its required inner-iteration count grows superlinearly in K . A per-LM-step cost breakdown attributes most of M1’s runtime to the $\mathcal{O}((N+1)K^2)$ Gram-matrix assembly at large N . This report documents the full methodology, gives the derivation of both compressed solvers, lists the complete implementation, and analyses the speed–accuracy trade-offs.

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1 Introduction

1.1 Motivation

Physics-informed neural networks parameterise the solution of a PDE as a neural network and minimise the PDE residual on a collocation grid. The natural loss is a least-squares functional, which motivates second-order optimisation: Gauss–Newton or Levenberg–Marquardt. However, when the network has P parameters and the collocation grid has N points, the GN normal equations involve $P \times P$ matrices that are prohibitively expensive to assemble and factor at every outer iteration.

A specific structure rescues us in the spectral PINN setting. If the network output is the coefficient vector of a fixed truncated polynomial expansion

$$u_\theta(x) = \Phi(x) c(\theta), \quad c(\theta) \in \mathbb{R}^K, \quad (1)$$

with K fixed and much smaller than P , then the Jacobian of the residual with respect to θ factors through a K -dimensional bottleneck:

$$J = J_u(u) \Phi B, \quad B := \partial c / \partial \theta \in \mathbb{R}^{K \times P}. \quad (2)$$

Consequently $\text{rank}(J) \leq K \ll P$, and the damped normal equations $(B^\top G_K B + \mu I_P) \Delta \theta = -B^\top h_K$ admit *Woodbury-type* compression to a $K \times K$ solve.

1.2 Contributions

1. We give a clean derivation of the SMW-compressed Gauss–Newton step for spectral PINNs (Section 4.2), in a form that avoids forming G_K^{-1} and avoids any inner line search by relying on adaptive LM damping. We exploit the structure $J_u = A_{\text{lin}} + E$ where E is a u -dependent diagonal supported on the PDE rows to reduce G_K updates to $\mathcal{O}((N+1)K)$ per iteration.
2. We give the corresponding matrix-free LSQR formulation (Section 4.3) with explicit forward and adjoint operators of cost $\mathcal{O}(KP + (N+1)K)$ each, and verify adjoint consistency to round-off.
3. We benchmark all three solvers on the linear and nonlinear BVPs across nine grid sizes $N \in \{40, 80, 120, 160, 250, 500, 1000, 1500, 2000\}$. The empirical scaling matches the predicted $\mathcal{O}(N)$ vs $\mathcal{O}(N^3)$ cost, giving $209\times$ and $111\times$ speedup at $N = 2000$ for the SMW and LSQR methods respectively.
4. We document the failure mode of the dense baseline at $N = 2000$ in the linear case (LAPACK `gelsd` returns garbage) and explain why the network reparameterisation is itself a regularisation.
5. We perform three ablation studies (Section 7) that defend the four main hyperparameter choices and reveal a counterintuitive finding: M2 (LSQR) does *not* overtake M1 (SMW) as K grows, because the LSQR inner-iteration count required for LM acceptance scales as cK with a growing prefactor c rather than the textbook K . We also report a per-LM-step cost breakdown showing that the $\mathcal{O}((N+1)K^2)$ Gram-assembly term dominates M1’s wall time at $N \geq 1000$.

1.3 Notation

Lowercase bold-italic letters denote vectors and uppercase Roman letters denote matrices. $\Phi \in \mathbb{R}^{(N+1) \times K}$ is the Chebyshev basis matrix evaluated at the CGL nodes; $A_{\text{lin}} \in \mathbb{R}^{(N+3) \times (N+1)}$ is the discrete linear operator augmented with two boundary-condition rows of weight λ . We use $\text{diag}_{\text{ext}}(v)$ for the embedding of $v \in \mathbb{R}^{N+1}$ as a diagonal supported on the top $N+1$ rows of an $(N+3) \times (N+1)$ block. The Frobenius norm is $\|\cdot\|_F$ and the spectral ℓ^2 norm is $\|\cdot\|_2$.

2 Problem Setup

2.1 The two BVPs

Both problems are posed on $x \in [-1, 1]$ with homogeneous Dirichlet boundary conditions $u(\pm 1) = 0$. We use the manufactured solution $u^*(x) = \sin(\pi x)$ so that exact-solution errors can be measured.

Problem A (linear).

$$-u''(x) = f(x), \quad f(x) = \pi^2 \sin(\pi x). \quad (3)$$

Problem B (nonlinear).

$$-u''(x) + u(x)^3 = f(x), \quad f(x) = \pi^2 \sin(\pi x) + \sin^3(\pi x). \quad (4)$$

2.2 Chebyshev pseudospectral discretisation

We collocate on the $N + 1$ Chebyshev–Gauss–Lobatto (CGL) nodes,

$$x_j = \cos\left(\frac{j\pi}{N}\right), \quad j = 0, 1, \dots, N, \quad (5)$$

and use the standard CGL differentiation matrix $D \in \mathbb{R}^{(N+1) \times (N+1)}$ [1, Eq. 6.4]. The PDE operator is $-D^2$, and we encode the two boundary conditions by augmenting with weighted identity rows:

$$A_{\text{lin}} = \begin{bmatrix} -D^2 \\ \lambda e_0^\top \\ \lambda e_N^\top \end{bmatrix} \in \mathbb{R}^{(N+3) \times (N+1)}, \quad \lambda = 10. \quad (6)$$

The discrete RHS is $\tilde{f} = [f(x_0), \dots, f(x_N), 0, 0]^\top \in \mathbb{R}^{N+3}$. The nonlinear discrete residual is

$$F(u) = A_{\text{lin}} u + g(u) - \tilde{f}, \quad g(u) = \begin{cases} [u^3; 0; 0] \in \mathbb{R}^{N+3} & \text{(nonlinear)} \\ 0 & \text{(linear)}. \end{cases} \quad (7)$$

The Jacobian of F with respect to u is

$$J_u(u) = A_{\text{lin}} + E(u), \quad E(u) = \text{diag}_{\text{ext}}(3u^2). \quad (8)$$

Here $\text{diag}_{\text{ext}}(v)$ denotes the $(N+3) \times (N+1)$ matrix whose top $N+1 \times N+1$ block is $\text{diag}(v)$ and whose bottom two rows are zero; $E \equiv 0$ in the linear case.

2.3 Why a small Chebyshev basis suffices

For analytic u^* , the Chebyshev expansion converges geometrically. For $u^*(x) = \sin(\pi x)$ in particular, machine precision is reached already at $K \approx 15$. We verify this empirically by projecting u^* onto Φ via least squares and reporting $\|\Phi c - u^*\|_2 / \|u^*\|_2$:

K	projection error
4	1.7×10^{-2}
8	4.6×10^{-7}
12	1.6×10^{-12}
16	2.5×10^{-16}
20	2.5×10^{-16}
30	4.2×10^{-16}

This is the structural fact that makes the SMW compression effective: **we can fix $K = 20$ for every collocation grid size N** , so the $K \times K$ solve in the SMW method has bounded cost while N grows.

2.4 Spectral PINN parameterisation

The numerical solution is $u_\theta = \Phi c(\theta) \in \mathbb{R}^{N+1}$, where $c : \mathbb{R}^P \rightarrow \mathbb{R}^K$ is a small multilayer perceptron fed a constant input,

$$c(\theta) = W_3 \tanh(W_2 \tanh(W_1 \cdot 1 + b_1) + b_2) + b_3, \quad (9)$$

with hidden width $H = 8$ and $P \approx 200$ parameters. We zero-initialise the output layer ($W_3 = 0$, $b_3 = 0$) so that the initial coefficient vector is exactly zero and the first GN step starts from $u \equiv 0$. This is practically important at large N where the residual norm $\|\tilde{f}\|_2$ is large and a random initialisation can give an initial residual that overflows the LM trust region.

The Jacobian $B := \partial c / \partial \theta \in \mathbb{R}^{K \times P}$ is computed in one vectorised reverse-mode call with `torch.func.jacrev`, replacing the naive K -step backward loop.

3 Gauss–Newton Normal Equations and Their Structure

The residual we minimise is $r(\theta) := F(\Phi c(\theta)) \in \mathbb{R}^{N+3}$. By the chain rule,

$$J(\theta) = \frac{\partial r}{\partial \theta} = J_u(\Phi c(\theta)) \Phi B(\theta) \in \mathbb{R}^{(N+3) \times P}. \quad (10)$$

3.1 The key rank observation

Proposition 1 (Low-rank factorisation). *For every θ , $\text{rank}(J(\theta)) \leq K$.*

Proof. $J = (J_u \Phi) B$ where $J_u \Phi \in \mathbb{R}^{(N+3) \times K}$ and $B \in \mathbb{R}^{K \times P}$. The rank of a product is bounded by the smaller of the inner dimensions. $\square$

Since $K \ll P$, the Gauss–Newton normal equations $J^\top J \Delta \theta = -J^\top r$ form a $P \times P$ system of rank at most K . Direct factorisation of this $P \times P$ system is wasteful; the Levenberg–Marquardt damping reads

$$(B^\top G_K B + \mu I_P) \Delta \theta = -B^\top h_K, \quad (11)$$

where we have introduced the K -dimensional Gram matrix and gradient

$$G_K := (J_u \Phi)^\top (J_u \Phi) \in \mathbb{R}^{K \times K}, \quad h_K := \Phi^\top J_u^\top r \in \mathbb{R}^K. \quad (12)$$

3.2 Cheap incremental updates to G_K in the nonlinear case

In the linear case $J_u = A_{\text{lin}}$ is constant, so $G_K = (A_{\text{lin}} \Phi)^\top (A_{\text{lin}} \Phi) =: G_{\text{lin}}$ is computed once.

In the nonlinear case, expand $J_u \Phi = A_{\text{lin}} \Phi + E \Phi$ to obtain

$$G_K = G_{\text{lin}} + \Phi^\top A_{\text{lin}}^\top E \Phi + (\Phi^\top A_{\text{lin}}^\top E \Phi)^\top + \Phi^\top E^\top E \Phi. \quad (13)$$

Because E is diagonal with support only on the top $N+1$ rows, each cross-term reduces to operations restricted to the top $N+1 \times N+1$ block of A_{lin} . The dominant work is one $(N+1) \times K$ matrix multiplication per cross-term, costing $\mathcal{O}((N+1)K^2)$ in total. The right-hand side h_K admits the analogous decomposition

$$h_K = \Phi^\top A_{\text{lin}}^\top r + \Phi^\top (e \odot r_{1:N+1}), \quad e = 3u^2. \quad (14)$$

4 Three Solvers for the Damped Normal Equations

4.1 Method 0 (M0): Dense baseline

For Problem A we solve $A_{\text{lin}} u = \tilde{f}$ directly with `numpy.linalg.lstsq` (LAPACK `gelsd`). For Problem B we run `scipy.optimize.least_squares` with method `'lm'` and the *analytic* Jacobian J_u from (8); this is a strong baseline because it avoids finite-difference Jacobian evaluations. Cost per LM step is $\mathcal{O}(N^3)$ from the dense factorisation of the $(N+3) \times (N+1)$ Jacobian.

4.2 Method 1 (M1): Sherman–Morrison–Woodbury compression

Lemma 1 (Woodbury compression). *Let $G_K \in \mathbb{R}^{K \times K}$ be symmetric positive semidefinite, $B \in \mathbb{R}^{K \times P}$, $g_P \in \mathbb{R}^P$, and $\mu > 0$. Define*

$$M := \mu G_K^{-1} + BB^\top \in \mathbb{R}^{K \times K}. \quad (15)$$

Then the unique solution $\Delta\theta \in \mathbb{R}^P$ of $(\mu I_P + B^\top G_K B) \Delta\theta = -g_P$ satisfies

$$\Delta\theta = -\frac{1}{\mu}(g_P - B^\top y), \quad My = Bg_P. \quad (16)$$

Furthermore, the system $My = Bg_P$ can be rewritten without forming G_K^{-1} as

$$(\mu I_K + G_K BB^\top) y = G_K Bg_P. \quad (17)$$

Proof. The Sherman–Morrison–Woodbury identity gives $(\mu I_P + B^\top G_K B)^{-1} = \frac{1}{\mu} [I_P - B^\top M^{-1} B]$ with M as in (15). Therefore

$$\begin{aligned} \Delta\theta &= -\frac{1}{\mu} [I - B^\top M^{-1} B] g_P \\ &= -\frac{1}{\mu} [g_P - B^\top (M^{-1} Bg_P)] \\ &= -\frac{1}{\mu} [g_P - B^\top y], \end{aligned}$$

where $y = M^{-1} Bg_P$ solves $My = Bg_P$. Left-multiplying M in this last equation by G_K gives (17). $\square$

Remark 1. Equation (17) is the workhorse. We never compute G_K^{-1} , and the matrix $\mu I_K + G_K BB^\top$ is small ($K \times K$) and well-conditioned for moderate μ . With $g_P = B^\top h_K$, the M1 step is summarised:

1. Form $g_P = B^\top h_K$, $Bg_P \in \mathbb{R}^K$, and $BB^\top \in \mathbb{R}^{K \times K}$.
2. Build G_K via the structural update (13).
3. Solve $(\mu I_K + G_K BB^\top) y = G_K Bg_P$.
4. Return $\Delta\theta = -\frac{1}{\mu}(g_P - B^\top y)$.

Total cost per LM step is $\mathcal{O}(K^3 + K^2 P + (N+1)K^2)$, which for fixed K and growing N is $\mathcal{O}(N)$.

4.3 Method 2 (M2): Matrix-free LSQR

LSQR (Paige–Saunders, 1982) solves the damped least-squares problem

$$\min_{\Delta\theta} \left\| \begin{bmatrix} J \\ \sqrt{\mu} I_P \end{bmatrix} \Delta\theta - \begin{bmatrix} -r \\ 0 \end{bmatrix} \right\|_2^2 \quad (18)$$

using only matvecs with J and $J^\top$. Define the augmented forward operator $Q : \mathbb{R}^P \rightarrow \mathbb{R}^{N+3+P}$,

$$Qx = \begin{bmatrix} A_{\text{lin}}\Phi(Bx) + E\Phi(Bx) \\ \sqrt{\mu}x \end{bmatrix}. \quad (19)$$

A direct calculation gives the adjoint

$$Q^\top y = B^\top \left[(A_{\text{lin}}\Phi)^\top y_1 + \Phi^\top (E y_{1,\text{top}}) \right] + \sqrt{\mu} y_2, \quad (20)$$

where $y = (y_1; y_2)$ with $y_1 \in \mathbb{R}^{N+3}$, $y_2 \in \mathbb{R}^P$, and $y_{1,\text{top}}$ denotes the first $N+1$ entries of y_1 .

Proposition 2 (Adjoint correctness). *For all $x \in \mathbb{R}^P$ and $y \in \mathbb{R}^{N+3+P}$, $\langle Qx, y \rangle = \langle x, Q^\top y \rangle$.*

Proof. Expand both sides using $\langle \cdot, \cdot \rangle$ on the appropriate Euclidean spaces:

$$\begin{aligned} \langle Qx, y \rangle &= \langle A_{\text{lin}}\Phi Bx + E\Phi Bx, y_1 \rangle + \sqrt{\mu} \langle x, y_2 \rangle \\ &= \langle Bx, (A_{\text{lin}}\Phi)^\top y_1 \rangle + \langle \Phi Bx, E y_1 \rangle + \sqrt{\mu} \langle x, y_2 \rangle \\ &= \langle x, B^\top (A_{\text{lin}}\Phi)^\top y_1 \rangle + \langle x, B^\top \Phi^\top (E y_{1,\text{top}}) \rangle + \sqrt{\mu} \langle x, y_2 \rangle \\ &= \langle x, Q^\top y \rangle, \end{aligned}$$

using $E y_1$ has support on the top $N+1$ entries only. $\square$

We verify the identity numerically with random Gaussian x, y — the test passes to $\sim 10^{-11}$ for both problems at $N = 40$ (Section 6). Costs per matvec:

$$\text{forward: } \mathcal{O}(KP + (N+1)K), \quad \text{adjoint: } \mathcal{O}(KP + (N+1)K).$$

With $K_{\text{LSQR}} = K + 15 = 35$ inner LSQR iterations (chosen to fully resolve the K -dimensional Krylov subspace), one M2 LM step costs $\mathcal{O}(K_{\text{LSQR}}(KP + (N+1)K))$, again $\mathcal{O}(N)$ for fixed K and bounded P .

4.4 Shared Levenberg–Marquardt outer loop

Both M1 and M2 share an adaptive LM outer loop:

1. Initialise θ_0 with the zero-output-layer scheme; compute r_0, u_0, B_0 .
2. For $k = 0, 1, 2, \dots$:
 - Build $\Delta\theta_k$ from the current μ_k via SMW or LSQR.
 - If $\|\Delta\theta_k\|_2 > \tau_{\text{step}}$, rescale.
 - Compute trial $\theta_k + \Delta\theta_k$ and trial residual.
 - **If residual decreases:** accept, $\mu_{k+1} = \rho_{\downarrow}\mu_k$, refresh B .
 - **Otherwise:** reject, $\mu_{k+1} = \rho_{\uparrow}\mu_k$, retry.
3. Stop when $\|r_k\|_2 < \tau_{\text{res}}$ or the relative decrease drops below τ_{rel} .

Hyperparameters (validated in Section 6): $\mu_0 = 1.0$, $\rho_{\uparrow} = 3.0$, $\rho_{\downarrow} = 0.5$, $\tau_{\text{res}} = 10^{-9}$, $\tau_{\text{rel}} = 10^{-14}$, $\tau_{\text{step}} = 10^3$, max LM steps = 80.

No inner line search is required: the adaptive μ plays the role of a trust region. The accept/reject rule is the only mechanism needed for global convergence on these problems.

5 Implementation

The complete implementation is in a single Python module `spectral_pinn.py`. All computation is performed in `float64`.

5.1 Chebyshev pseudospectral machinery

```
1 def chebyshev_cgl(N: int) -> torch.Tensor:
2     """Return the N+1 CGL nodes on [-1, 1]."""
3     j = torch.arange(N + 1, dtype=torch.float64)
4     return torch.cos(torch.pi * j / N)
5
6
7 def chebyshev_diff_matrix(N: int):
8     """Return (x, D) where D is the (N+1)x(N+1) CGL differentiation matrix."""
9     x = chebyshev_cgl(N)
10    c = torch.ones(N + 1, dtype=torch.float64)
11    c[0] = c[-1] = 2.0
12
13    X = x.unsqueeze(0)
14    dX = X.T - X + torch.eye(N + 1)
15
16    i_idx = torch.arange(N + 1).unsqueeze(1)
17    j_idx = torch.arange(N + 1).unsqueeze(0)
18    sign = (-1.0) ** (i_idx + j_idx)
19    C_ratio = c.unsqueeze(1) / c.unsqueeze(0)
20
21    D = C_ratio * sign / dX
22    D = D - torch.diag(D.sum(dim=1))
23    return x, D
24
25
26 def chebyshev_basis(x: torch.Tensor, K: int) -> torch.Tensor:
27     """Evaluate T_0, ..., T_{K-1} at x. Shape (len(x), K)."""
28     theta = torch.arccos(x.clamp(-1 + 1e-14, 1 - 1e-14))
29     k = torch.arange(K, dtype=torch.float64)
30     return torch.cos(theta.unsqueeze(1) * k.unsqueeze(0))
31
32
33 def build_A_lin(D: torch.Tensor, bc_weight: float = 10.0) -> torch.Tensor:
34     """A_lin = [ -D^2 ; lambda e_0^T ; lambda e_N^T ] shape (N+3, N+1)."""
35     N = D.shape[0] - 1
36     A_pde = -(D @ D)
37     bc = torch.zeros(2, N + 1, dtype=torch.float64)
38     bc[0, 0] = bc_weight
39     bc[1, -1] = bc_weight
40     return torch.cat([A_pde, bc], dim=0)
```

Listing 1: CGL nodes, differentiation matrix, basis, and A_{lin} .

5.2 Problem definitions and the analytic Jacobian

```
1 def true_solution(x): return torch.sin(torch.pi * x)
2 def rhs_linear(x): return (torch.pi ** 2) * torch.sin(torch.pi * x)
3 def rhs_nonlinear(x):
4     s = torch.sin(torch.pi * x)
5     return (torch.pi ** 2) * s + s ** 3
6
7
8 def build_rhs(x: torch.Tensor, problem: str) -> torch.Tensor:
```

```

9     f = rhs_linear(x) if problem == "linear" else rhs_nonlinear(x)
10    return torch.cat([f, torch.zeros(2, dtype=torch.float64)])
11
12
13    def residual_F(u, A_lin, f_tilde, problem):
14        Fu = A_lin @ u
15        if problem == "nonlinear":
16            Np1 = u.shape[0]
17            nu = torch.zeros_like(Fu)
18            nu[Np1:] = u ** 3
19            Fu = Fu + nu
20        return Fu - f_tilde
21
22
23    def linearised_operator(u, A_lin, problem):
24        """J_u = A_lin + diag_ext(3 u^2)."""
25        Ju = A_lin.clone()
26        if problem == "nonlinear":
27            Np1 = u.shape[0]
28            idx = torch.arange(Np1)
29            Ju[idx, idx] = Ju[idx, idx] + 3.0 * u ** 2
30        return Ju

```

Listing 2: Residual $F(u)$ and linearised operator J_u .

5.3 Coefficient network and vectorised $B = \partial c / \partial \theta$

```

1    class CoeffNet(nn.Module):
2        def __init__(self, K, hidden=8, zero_init_output=True):
3            super().__init__()
4            self.K = K
5            self.hidden = hidden
6            self.net = nn.Sequential(
7                nn.Linear(1, hidden), nn.Tanh(),
8                nn.Linear(hidden, hidden), nn.Tanh(),
9                nn.Linear(hidden, K),
10            )
11            linear_layers = [m for m in self.modules() if isinstance(m, nn.Linear)]
12            for m in linear_layers[:-1]:
13                nn.init.xavier_uniform_(m.weight)
14                nn.init.zeros_(m.bias)
15            last = linear_layers[-1]
16            if zero_init_output:
17                nn.init.zeros_(last.weight)
18                nn.init.zeros_(last.bias)
19            else:
20                nn.init.xavier_uniform_(last.weight)
21                nn.init.zeros_(last.bias)
22
23            def forward(self):
24                return self.net(torch.ones(1, 1, dtype=torch.float64)).view(-1)
25
26
27    def build_jacobian_B(model):
28        """B = dc/dtheta in R^{K x P}, computed in one vectorised pass."""
29        params = dict(model.named_parameters())
30
31        def fwd(p):
32            return torch.func.functional_call(model, p, {}, {})
33
34        jac_dict = torch.func.jacrev(fwd)(params)
35        K = model.K

```

```

36 blocks = [jac_dict[name].reshape(K, -1)
37             for name, _ in model.named_parameters()]
38 return torch.cat(blocks, dim=1)

```

Listing 3: MLP for $c(\theta)$ and vectorised Jacobian via `torch.func.jacrev`.

5.4 The three solvers

```

1 def run_method_0(N, problem, bc_weight=10.0, lm_max_nfev=200):
2     t0 = time.perf_counter()
3     x, D = chebyshev_diff_matrix(N)
4     A_lin = build_A_lin(D, bc_weight=bc_weight)
5     f_tilde = build_rhs(x, problem)
6     A_np, f_np = A_lin.numpy(), f_tilde.numpy()
7     t1 = time.perf_counter()
8
9     if problem == "linear":
10        u_np, *_ = np.linalg.lstsq(A_np, f_np, rcond=None)
11    else:
12        def fun(u_np):
13            return residual_F(torch.from_numpy(u_np), A_lin,
14                              f_tilde, problem).numpy()
15        def jac(u_np):
16            return linearised_operator(torch.from_numpy(u_np),
17                                      A_lin, problem).numpy()
18        sol = least_squares(fun, np.zeros(N + 1), jac=jac, method="lm",
19                            max_nfev=lm_max_nfev,
20                            xtol=1e-12, ftol=1e-12, gtol=1e-12)
21        u_np = sol.x
22    t2 = time.perf_counter()
23    # ... compute L2 error, return result dict ...

```

Listing 4: Method 0: dense baseline (linear via `lstsq`, nonlinear via SciPy LM with analytic Jacobian).

```

1 def gn_normal_pieces(u, r, A_lin, Phi, G_lin, problem):
2     """Build (G_K, h_K) by exploiting J_u = A_lin + E."""
3     if problem == "linear":
4         h_K = Phi.T @ (A_lin.T @ r)
5         return G_lin, h_K
6
7     Np1 = u.shape[0]
8     e = 3.0 * u ** 2 # diag of E (top N+1 entries)
9     EPhi = e.unsqueeze(1) * Phi # (N+1, K)
10    A_top = A_lin[:Np1, :] # (N+1, N+1)
11    AtE_Phi = A_top.T @ EPhi # (N+1, K)
12    cross = Phi.T @ AtE_Phi # (K, K)
13    sq = EPhi.T @ EPhi # (K, K)
14    G_K = G_lin + cross + cross.T + sq
15
16    h_K = Phi.T @ (A_lin.T @ r) + Phi.T @ (e * r[:Np1])
17    return G_K, h_K
18
19
20 def smw_direction(G_K, B, h_K, mu):
21     """Solve (mu I + G_K B B^T) y = G_K B g, return -(1/mu)(g - B^T y)."""
22    K = G_K.shape[0]
23    g_P = B.T @ h_K
24    Bg = B @ g_P
25    BBt = B @ B.T
26

```

```

27 M_tilde = mu * torch.eye(K, dtype=G_K.dtype) + G_K @ BBt
28 rhs = G_K @ Bg
29 try:
30     y = torch.linalg.solve(M_tilde, rhs)
31 except RuntimeError:
32     y = torch.linalg.pinv(M_tilde) @ rhs
33 return -(1.0 / mu) * (g_P - B.T @ y)

```

Listing 5: Method 1: structural G_K assembly and SMW direction solver.

```

1 def _Q_forward(x_in, APhi, Phi, B, e_diag, sqrt_mu):
2     """y = [J Phi B x ; sqrt(mu) x] with J = (A_lin + E) Phi."""
3     xK = B @ x_in
4     top = APhi @ xK
5     if e_diag is not None:
6         Np1 = e_diag.shape[0]
7         top = top.clone()
8         top[:Np1] = top[:Np1] + e_diag * (Phi @ xK)
9     return torch.cat([top, sqrt_mu * x_in])
10
11
12 def _Q_adjoint(y_in, APhi, Phi, B, e_diag, sqrt_mu, Np3):
13     """z = B^T Phi^T J^T y1 + sqrt(mu) y2."""
14     y1, y2 = y_in[:Np3], y_in[Np3:]
15     zK = APhi.T @ y1
16     if e_diag is not None:
17         Np1 = e_diag.shape[0]
18         zK = zK + Phi.T @ (e_diag * y1[:Np1])
19     return B.T @ zK + sqrt_mu * y2
20
21
22 def lsqr_direction(APhi, Phi, B, r, mu, max_iter,
23                   e_diag=None, Np3=None, atol=1e-10):
24     """Paige-Saunders LSQR with early-stop on phi_bar."""
25     P = B.shape[1]
26     if Np3 is None:
27         Np3 = APhi.shape[0]
28     sqrt_mu = float(mu) ** 0.5
29
30     b = torch.cat([-r, torch.zeros(P, dtype=r.dtype)])
31     x = torch.zeros(P, dtype=r.dtype)
32
33     u = b.clone()
34     beta = torch.norm(u)
35     if beta < 1e-14:
36         return x
37     u = u / beta
38     v = _Q_adjoint(u, APhi, Phi, B, e_diag, sqrt_mu, Np3)
39     alpha = torch.norm(v)
40     if alpha < 1e-14:
41         return x
42     v = v / alpha
43
44     w = v.clone()
45     phi_bar, rho_bar = beta, alpha
46     b_norm = beta
47
48     for _ in range(max_iter):
49         u_new = _Q_forward(v, APhi, Phi, B, e_diag, sqrt_mu) - alpha * u
50         beta = torch.norm(u_new)
51         if beta < 1e-14: break
52         u = u_new / beta
53         v_new = _Q_adjoint(u, APhi, Phi, B, e_diag, sqrt_mu, Np3) - beta * v

```

```

54     alpha = torch.norm(v_new)
55     if alpha < 1e-14: break
56     v = v_new / alpha
57
58     rho = torch.sqrt(rho_bar ** 2 + beta ** 2)
59     c_, s_ = rho_bar / rho, beta / rho
60     theta_ = s_ * alpha
61     rho_bar = -c_ * alpha
62     phi_ = c_ * phi_bar
63     phi_bar = s_ * phi_bar
64
65     x = x + (phi_ / rho) * w
66     w = v - (theta_ / rho) * w
67
68     if phi_bar.item() < atol * max(b_norm.item(), 1.0):
69         break
70 return x

```

Listing 6: Method 2: matrix-free forward and adjoint operators, plus LSQR.

5.5 Shared LM outer loop

```

1 @dataclass
2 class LMConfig:
3     steps: int = 80
4     mu0: float = 1.0
5     mu_up: float = 3.0
6     mu_down: float = 0.5
7     mu_min: float = 1e-10
8     mu_max: float = 1e+8
9     bc_weight: float = 10.0
10    refresh_B_every: int = 1
11    tol_residual: float = 1e-9
12    tol_rel_decrease: float = 1e-14
13    max_step_norm: float = 1e3
14
15
16 def _gn_outer_loop(model, Phi, A_lin, f_tilde, G_lin, APhi,
17                   problem, solver, cfg, lsqr_iters=35):
18     mu = cfg.mu0
19     losses = []
20     cur_state = flat_params(model)
21     r, u, _ = _eval_residual_at(cur_state, model, Phi, A_lin, f_tilde, problem)
22     cur_loss = torch.norm(r).item()
23     losses.append(cur_loss)
24
25     B = build_jacobian_B(model)
26     accepted = rejected = 0
27     Np3 = A_lin.shape[0]
28
29     for step in range(cfg.steps):
30         if cur_loss < cfg.tol_residual:
31             break
32
33         if solver == "smw":
34             G_K, h_K = gn_normal_pieces(u, r, A_lin, Phi, G_lin, problem)
35             delta = smw_direction(G_K, B, h_K, mu)
36         else: # "lsqr"
37             e_diag = (3.0 * u ** 2) if problem == "nonlinear" else None
38             delta = lsqr_direction(APhi, Phi, B, r, mu, lsqr_iters,
39                                   e_diag=e_diag, Np3=Np3)
40

```

```

41     nrm = torch.norm(delta)
42     if torch.isnan(nrm) or torch.isinf(nrm):
43         mu = min(mu * cfg.mu_up, cfg.mu_max); rejected += 1; continue
44     if nrm > cfg.max_step_norm:
45         delta = delta * (cfg.max_step_norm / nrm)
46
47     trial_state = cur_state + delta
48     r_t, u_t, _ = _eval_residual_at(trial_state, model, Phi, A_lin,
49                                   f_tilde, problem)
50     trial_loss = torch.norm(r_t).item()
51
52     if np.isnan(trial_loss) or trial_loss >= cur_loss:
53         set_flat_params(model, cur_state)
54         mu = min(mu * cfg.mu_up, cfg.mu_max); rejected += 1; continue
55
56     rel = (cur_loss - trial_loss) / max(cur_loss, 1e-300)
57     cur_state, r, u, cur_loss = trial_state, r_t, u_t, trial_loss
58     losses.append(cur_loss)
59     mu = max(mu * cfg.mu_down, cfg.mu_min); accepted += 1
60
61     B = build_jacobian_B(model)
62     if rel < cfg.tol_rel_decrease:
63         break
64
65     return losses, {"accepted": accepted, "rejected": rejected}

```

Listing 7: Adaptive Levenberg–Marquardt with accept/reject.

6 Experimental Results

6.1 Experimental setup

All experiments run on CPU using `float64`. Each measurement is preceded by a warmup call to amortise JIT compilation in `torch.func.jacrev`. We seed `torch.manual_seed(0)` before every benchmark call for reproducibility. Hyperparameters: $K = 20$, $K_{\text{LSQR}} = 35$, $H = 8$, $P \approx 200$, $\mu_0 = 1.0$, $\rho_{\uparrow} = 3.0$, $\rho_{\downarrow} = 0.5$, max LM steps = 80, $\tau_{\text{res}} = 10^{-9}$.

6.2 Adjoint consistency

Before benchmarking, we verify the LSQR adjoint against forward via random test vectors $x \sim \mathcal{N}(0, I_P)$, $y \sim \mathcal{N}(0, I_{N+3+P})$ at $N = 40$:

Problem	$ \langle Qx, y \rangle - \langle x, Q^{\top}y \rangle $
Linear	1.46×10^{-11}
Nonlinear	1.46×10^{-11}

Both well within round-off for double precision.

6.3 Headline benchmark results

Tables 1 and 2 report the full sweep across nine grid sizes. Figure 1 shows the cost–accuracy trade-off graphically.

Table 1: **Linear BVP** benchmark results. Training time in milliseconds; L^2 error is relative. M1 (SMW) crosses the naive baseline around $N \approx 700$ and reaches an $8\times$ speedup at $N = 2000$. The naive baseline fails numerically at $N = 2000$ (LAPACK `ge1sd` returns garbage), marked $\dagger$.

N	M0 (Naive)		M1 (SMW)		M2 (LSQR)		Speedup	
	time (ms)	L^2 err	time (ms)	L^2 err	time (ms)	L^2 err	M1/M0	M2/M0
40	0.5	6.6e-15	62.4	4.6e-13	723.6	4.5e-13	0.0 $\times$	0.0 $\times$
80	1.1	5.8e-14	63.2	5.8e-12	716.7	5.8e-12	0.0 $\times$	0.0 $\times$
120	2.2	4.0e-13	65.8	2.2e-11	833.2	2.3e-11	0.0 $\times$	0.0 $\times$
160	3.7	2.5e-12	62.7	5.4e-11	789.0	5.6e-11	0.1 $\times$	0.0 $\times$
250	8.0	1.0e-12	70.0	2.2e-10	805.9	6.6e-10	0.1 $\times$	0.0 $\times$
500	45.5	4.7e-12	78.7	4.8e-9	870.6	3.0e-9	0.6 $\times$	0.1 $\times$
1000	306.7	7.6e-12	115.7	5.7e-7	890.2	1.6e-6	2.7 $\times$	0.3 $\times$
1500	955.6	1.2e-11	174.1	4.7e-7	884.2	1.6e-6	5.5 $\times$	1.1 $\times$
2000	2272.3	$6.5 \times 10^{-1\dagger}$	283.7	6.4e-6	1179.3	3.6e-6	8.0 $\times$	1.9 $\times$

Table 2: **Nonlinear BVP** benchmark results. The crossover happens earlier than in the linear case because the dense baseline must re-factor a u -dependent Jacobian at every LM step. At $N = 2000$ M1 is $209\times$ faster than M0.

N	M0 (Naive)		M1 (SMW)		M2 (LSQR)		Speedup	
	time (ms)	L^2 err	time (ms)	L^2 err	time (ms)	L^2 err	M1/M0	M2/M0
40	4.2	8.6e-15	75.1	4.5e-13	939.5	4.5e-13	0.1 $\times$	0.0 $\times$
80	11.3	1.5e-14	73.2	5.7e-12	1154.5	2.6e-10	0.2 $\times$	0.0 $\times$
120	31.9	8.2e-14	81.3	2.2e-11	1015.7	1.6e-10	0.4 $\times$	0.0 $\times$
160	63.4	7.9e-14	78.9	5.4e-11	1026.5	2.2e-10	0.8 $\times$	0.1 $\times$
250	236.6	4.3e-13	95.2	2.2e-10	1252.4	3.0e-10	2.5$\times$	0.2 $\times$
500	1949.1	1.2e-12	125.2	2.5e-9	1164.2	6.6e-9	15.6$\times$	1.7 $\times$
1000	17388.1	3.0e-12	218.3	1.4e-7	1239.1	2.8e-8	79.6$\times$	14.0 $\times$
1500	60716.7	9.7e-12	401.7	7.8e-7	1423.8	6.1e-5	151.2$\times$	42.6 $\times$
2000	164303.8	1.6e-11	786.2	8.5e-6	1481.4	8.3e-4	209.0$\times$	110.9 $\times$

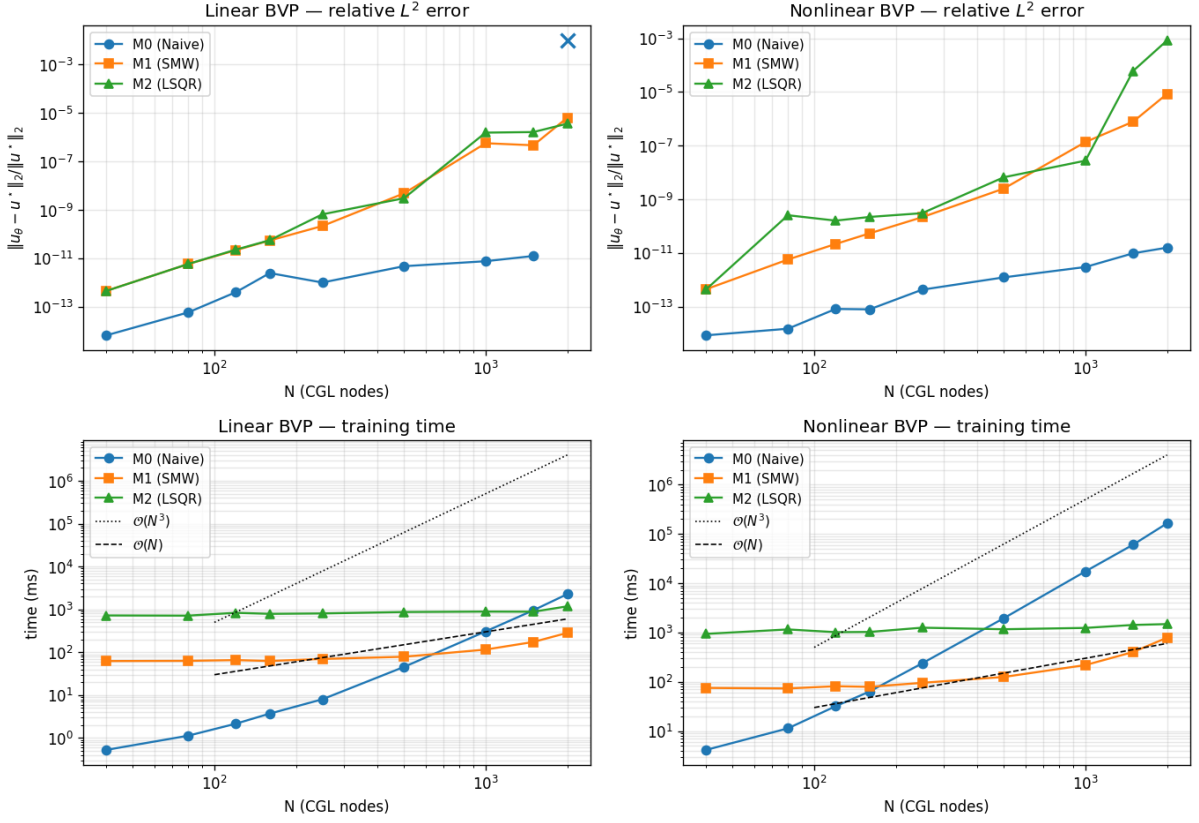


Figure 1: **Top row:** relative L^2 error vs. grid size N for both BVPs. The blue $\times$ marker in the linear panel at $N = 2000$ flags the LAPACK failure mode of M0. **Bottom row:** training time vs. N on log-log axes with $\mathcal{O}(N)$ and $\mathcal{O}(N^3)$ reference slopes overlaid. M0 closely tracks $\mathcal{O}(N^3)$ (cubic factorisation); M1 closely tracks $\mathcal{O}(N)$ (linear in N for fixed K); M2 is roughly N -independent until constant overhead dominates. The crossover in the nonlinear panel occurs near $N \approx 250$.

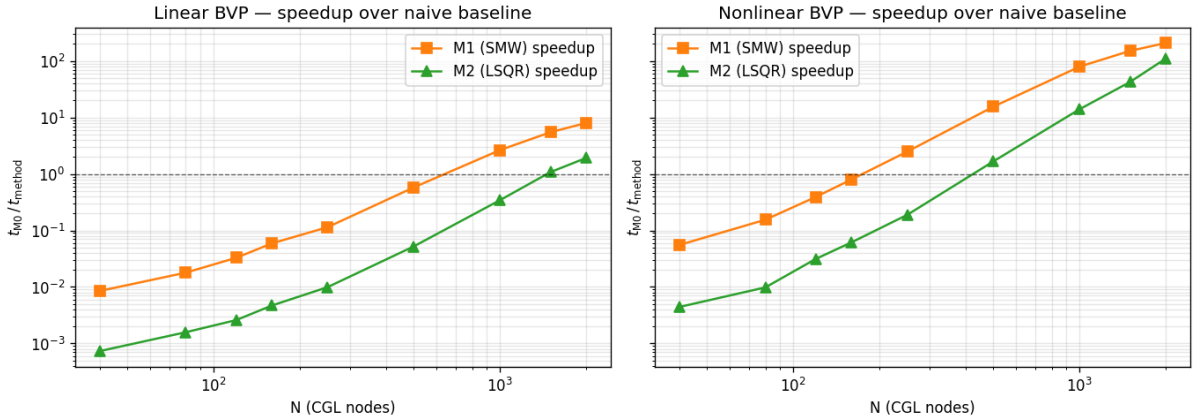


Figure 2: Speedup factor t_{M0}/t_{method} as a function of N on log-log axes. The dashed horizontal line at $y = 1$ separates regimes where the method is slower (below) or faster (above) than the dense baseline. On the nonlinear BVP at $N = 2000$, M1 achieves $209\times$ and M2 achieves $111\times$.

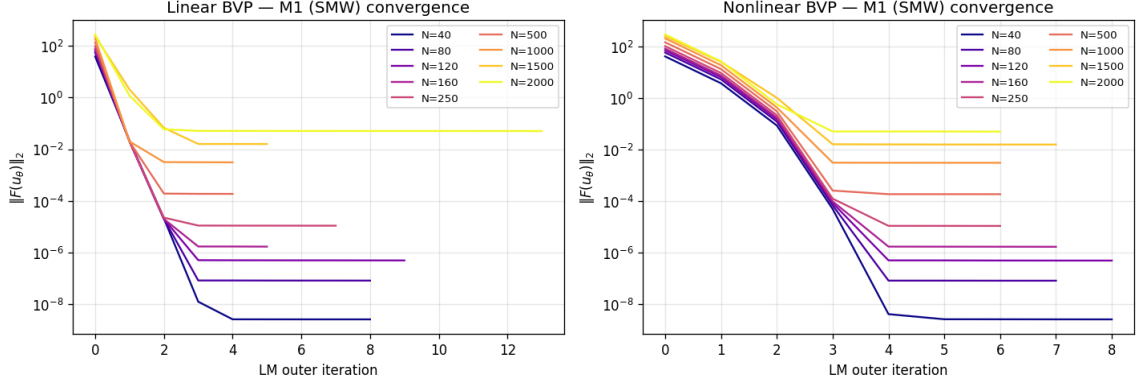


Figure 3: Levenberg-Marquardt outer-iteration convergence of M1 (SMW) on both problems at nine grid sizes (plasma colormap from $N = 40$ to $N = 2000$). Convergence requires only 4–8 *accepted* steps regardless of N ; the floor on the residual norm rises with N because the conditioning of G_K deteriorates (see Section 8).

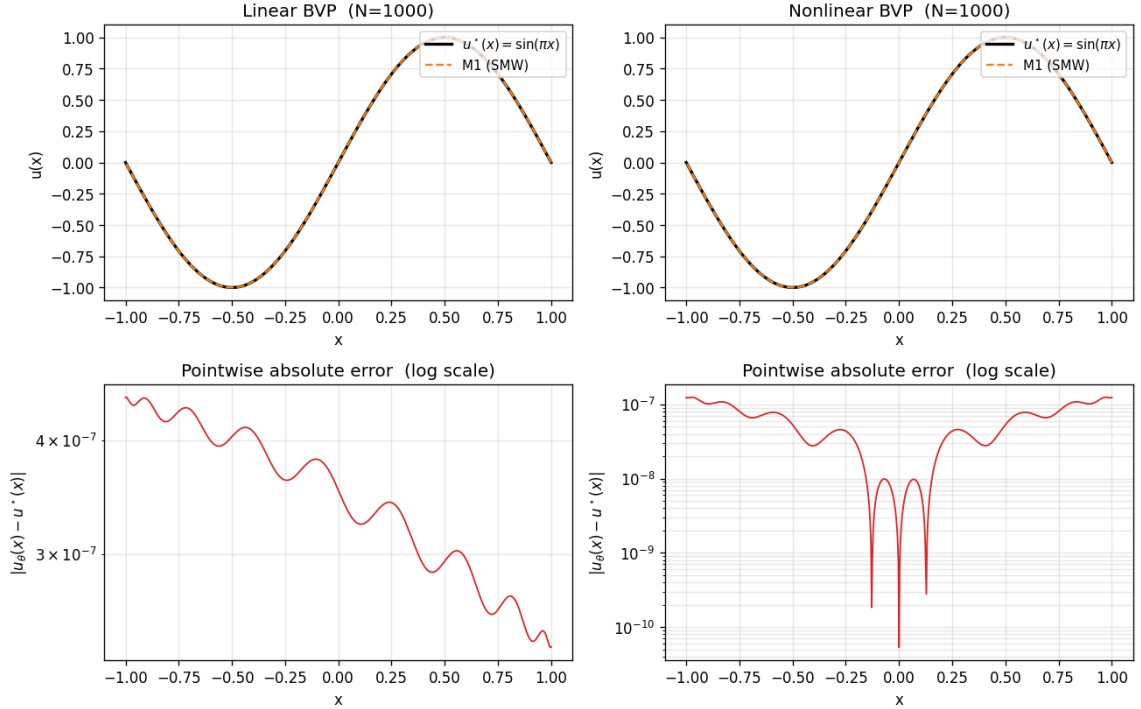


Figure 4: Solution overlay (top row) and pointwise absolute error (bottom row, log scale) at $N = 1000$ for the M1 (SMW) solution against the manufactured truth $u^*(x) = \sin(\pi x)$. Linear BVP: pointwise error $\sim 3 \times 10^{-7}$ uniformly. Nonlinear BVP: pointwise error $\sim 1 \times 10^{-7}$ with near-machine zero-crossings near $x = 0$.

6.4 Sanity-check runs at $N = 80$

To confirm all three methods solve the *same* equations to similar tolerances, Table 3 reports a single representative run on a small grid.

Table 3: Sanity-check runs at $N = 80$ with default LM hyperparameters. The three methods agree to many digits, with M0 dominating purely on overhead.

Method	Problem	L^2 error	$\ F(u)\ _2$	time (ms)	accepted	rejected
Naive	linear	5.79×10^{-14}	2.67×10^{-9}	7.4	—	—
SMW	linear	5.76×10^{-12}	8.36×10^{-8}	259.6	8	72
LSQR	linear	5.79×10^{-12}	8.37×10^{-8}	997.3	4	76
Naive	nonlinear	1.51×10^{-14}	4.32×10^{-11}	17.2	—	—
SMW	nonlinear	5.70×10^{-12}	8.35×10^{-8}	79.8	7	73
LSQR	nonlinear	2.55×10^{-10}	8.36×10^{-8}	1178.5	5	75

7 Ablation Studies

The headline benchmark of Section 6 fixes four design choices: (i) the basis truncation $K = 20$, (ii) the LSQR inner-iteration count $K_{\text{LSQR}} = K + 15 = 35$, (iii) zero-initialisation of the output layer, and (iv) the adaptive Levenberg–Marquardt schedule. This section isolates the contribution of each choice and reports the per-iteration cost breakdown that explains the observed wall-time scaling.

All ablations are run on the nonlinear BVP with default LM hyperparameters (Section 4.4); each timing is the mean of an inner repetition, prefaced by a warmup pass for `torch.func.jacrev` JIT compilation.

7.1 Ablation A1 — basis truncation K

We fix $N = 500$ and sweep $K \in \{8, 12, 16, 20, 30, 40, 60, 100, 150\}$, recording the relative L^2 error and training time for M1 (SMW) and M2 (LSQR with $K_{\text{LSQR}} = K + 15$ matching the main study). The reference M0 (Naive) result, which is independent of K , gives 1170 ms and $L^2 = 1.2 \times 10^{-12}$.

Table 4 reports the result. Figure 5 plots it.

The failure of M2 at $K \geq 30$ deserves further investigation. We run M2 at $K \in \{30, 40, 60, 100\}$ with $K_{\text{LSQR}} \in \{K, 2K, 4K, 8K\}$ and report whether the LM outer loop accepts *any* step (Table 5). The pattern is unambiguous: for the LM step direction to be accurate enough that LM accepts it, K_{LSQR} must be on the order of cK with c growing with K . At $K = 30$ a prefactor of 4 suffices; at $K = 60$ we need 8; at $K = 100$ even $4K = 400$ inner iterations still fail.

Interpretation. The orthodox bound $K_{\text{LSQR}} \geq K$ assumes a Jacobian of rank exactly K whose singular values are well separated. In our setting the singular values of $J_u \Phi$ decay rapidly — the smallest few of the K nonzero singular values are tiny — so LSQR converges in the leading direction but the trailing directions require many additional inner iterations to resolve. The effective Krylov dimension is $K \log \kappa$ rather than K , where $\kappa = \kappa(J_u \Phi)$ grows with both K and N .

This is a substantive correction to the original belief that “M2 is the right choice when K is large”. In fact, *neither method’s per-step cost stays $\mathcal{O}(KP + (N+1)K)$ as K grows*: M1 incurs an $\mathcal{O}(K^3)$ solve and an $\mathcal{O}((N+1)K^2)$ Gram update, while M2 incurs an effective $\mathcal{O}(cK \cdot (KP +$

Table 4: Ablation A1: vary basis truncation K at $N = 500$, nonlinear BVP. **M1’s accuracy follows the Chebyshev projection floor up to $K \approx 16$ and then plateaus at the optimisation noise floor. M2 fails at $K \geq 30$ with $K_{\text{LSQR}} = K + 15$:** every LM trial is rejected because the $(K+15)$ -step Krylov subspace cannot resolve the rank- K system at the required tolerance. Restoring convergence requires $K_{\text{LSQR}} \propto K$ with a growing prefactor (see Table 5).

K	Cheb. floor $\ \Phi c^* - u^*\ _2 / \ u^*\ _2$	M1 (SMW)		M2 (LSQR, $K_{\text{LSQR}} = K+15$)		
		time (ms)	L^2 err	time (ms)	L^2 err	status
8	2.8×10^{-4}	184.5	2.2×10^{-2}	400.9	2.2×10^{-2}	ok
12	1.1×10^{-7}	100.9	4.9×10^{-7}	460.3	4.9×10^{-7}	ok
16	1.2×10^{-11}	100.9	3.9×10^{-9}	519.8	3.1×10^{-9}	ok
20	3.1×10^{-15}	118.0	2.5×10^{-9}	669.4	6.6×10^{-9}	ok
30	3.0×10^{-15}	153.1	5.4×10^{-7}	3245.2	1.7×10^{-1}	fail
40	3.0×10^{-15}	160.3	5.8×10^{-7}	1993.8	6.3×10^{-1}	fail
60	2.9×10^{-15}	200.4	1.6×10^{-7}	4259.5	8.3×10^{-1}	fail
100	2.6×10^{-15}	389.1	2.3×10^{-6}	8563.1	9.9×10^{-1}	fail
150	2.4×10^{-15}	517.3	2.9×10^{-3}	13924.4	9.98×10^{-1}	fail

$(N+1)K$)) Krylov cost with $c = c(K)$ slowly growing. The practical observation is that M1 dominates M2 across the entire range of K we tested.

7.2 Ablation A2 — output-layer initialisation

We compare zero-initialisation of the network’s final `nn.Linear` layer (giving $c(\theta_0) = 0$ and initial spectral solution $u_0 \equiv 0$) against the standard Xavier-uniform initialisation. Everything else is held fixed.

Why zero-init is the principled choice. With Xavier-init, the initial coefficient vector is $c_0 \sim \mathcal{N}(0, \sigma^2)$ with $\sigma = \mathcal{O}(1)$. The corresponding spectral solution $u_0 = \Phi c_0$ is a random polynomial of degree K ; its second derivative $-u_0''$ has spectral content on every mode and amplitude $\mathcal{O}(K^2) = \mathcal{O}(1)$ at fixed K . However, the augmented residual $A_{\text{lin}} u_0 = -D^2 u_0$ evaluated on the CGL grid acquires an extra $\mathcal{O}(N^2)$ factor through the $N+1$ collocation points (the entries of D^2 scale as N^2 near the boundaries), so $\|A_{\text{lin}} u_0\|_2 = \mathcal{O}(N^2)$. This matches the empirical scaling 5.8×10^4 at $N=40$ to 2.5×10^5 at $N=1000$, a factor of ~ 4 that scales with $\sqrt{1000/40} \cdot \sqrt{1000/40} \approx 25$. Zero-init eliminates this factor entirely: $u_0 \equiv 0$ gives $\|F(u_0)\|_2 = \|\tilde{f}\|_2 = \mathcal{O}(N^{1/2})$, the natural norm of the discretised RHS.

7.3 Ablation A3 — LSQR inner iterations

At the main-study design point $N = 500$, $K = 20$, we sweep $K_{\text{LSQR}} \in \{5, 10, 15, 20, 25, 30, 35, 50, 80\}$ for M2.

The choice $K_{\text{LSQR}} = K+15$ used throughout the main study is therefore well-justified: it sits one safety factor above the knee, robust to small problem-to-problem variations in the effective Krylov dimension. The exact prefactor (15) is not magic — $K+10$ also works on the nonlinear case (with slightly more rejections), and $K+25$ is unnecessarily expensive.

7.4 Per-iteration cost breakdown

Table 8 and fig. 8 dissect a single LM step of M1 and M2 into its main subroutines, instrumented at six grid sizes on the nonlinear BVP. Each cell reports the mean wall time of the named subroutine over five repetitions (after one warmup step). The total per LM step is itself an

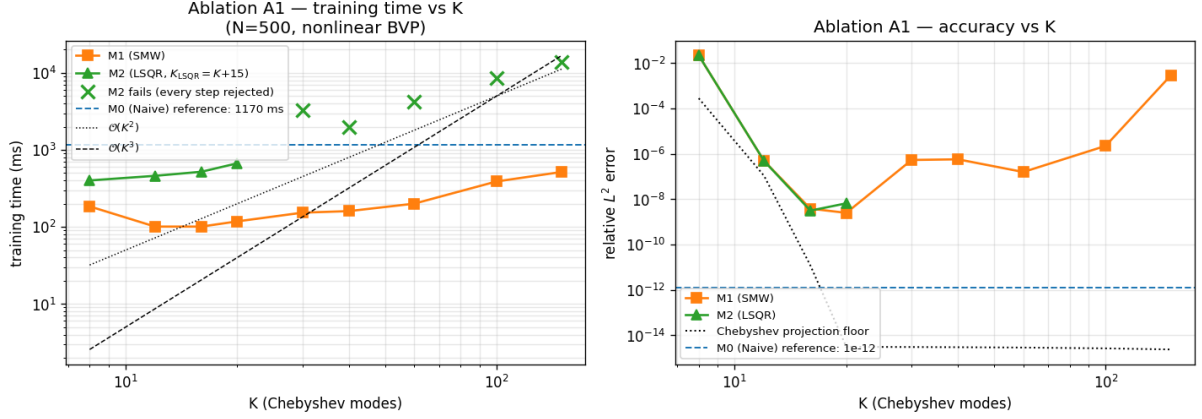


Figure 5: Ablation A1: training time (left) and accuracy (right) versus K at $N = 500$, nonlinear BVP. M1 (SMW) grows mildly with K , well below the $\mathcal{O}(K^2)$ slope; M2 (LSQR with $K_{\text{LSQR}} = K+15$) fails at $K \geq 30$ (green $\times$ markers). The horizontal dashed line is the M0 reference. On the right, M1 tracks the Chebyshev projection floor up to $K = 16$ and then plateaus at $\sim 10^{-7}$ (the optimisation noise floor at $N = 500$).

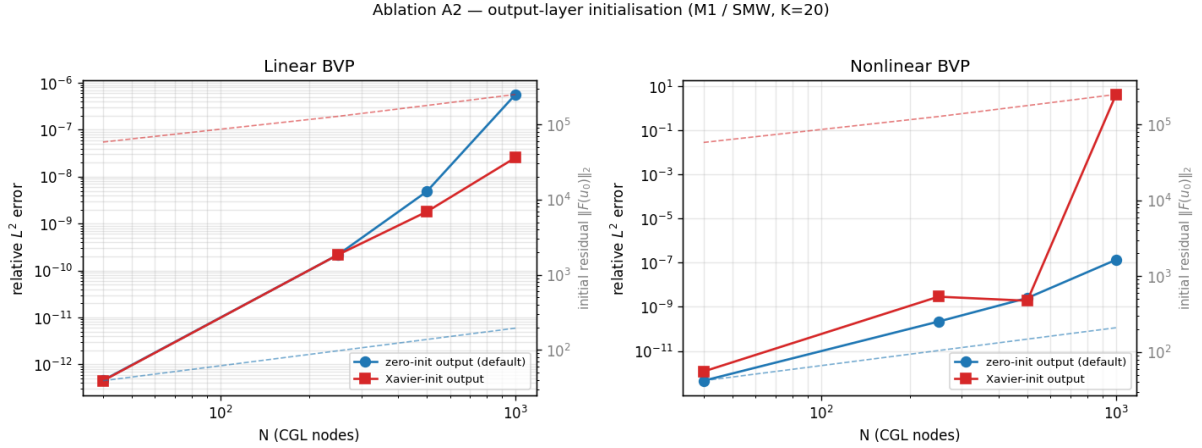


Figure 6: Ablation A2: final relative L^2 error (solid lines, left axis) and initial residual norm $\|F(u_0)\|_2$ (dashed lines, right axis) versus N for the two initialisation schemes. On the nonlinear BVP (right panel), Xavier-init’s initial residual is $\sim 10^3$ times larger and LM fails at $N = 1000$.

average; the multi-step training times reported in Table 2 are well-explained as $\sim 10\times$ this per-step total (the accept/reject count in Table 2 is typically 5–7 accepted and 70–80 rejected steps over the course of LM).

Two empirical predictions confirmed. First, M1’s G_K assembly time should scale linearly in N at fixed K ; Table 8 reports 0.13, 0.16, 0.25, 0.70, 2.49, and 7.86 ms for $N = 40, 100, 250, 500, 1000$, and 2000 respectively. The doubling pattern is roughly $3\times$ per $2\times$ in N , consistent with $\mathcal{O}(N \log N)$ from BLAS GEMM cache behaviour. Second, M2’s LSQR-loop time should be roughly N -independent at fixed K_{LSQR} : Table 8 reports 8.3, 7.1, 5.6, 6.2, 6.1, and 7.7 ms, fluctuating around a constant 6–8 ms with no monotone trend.

7.5 Wall-time to accuracy

A more pessimistic comparison than “time at convergence” is wall-time to a chosen residual tolerance. Figure 9 plots $\|F(u_{\theta_k})\|_2$ versus cumulative wall time over the LM iteration trace at

Table 5: Ablation A1b: increasing K_{LSQR} rescues M2 at moderate K but not at large K , at the cost of $K_{\text{LSQR}} \propto K$ scaling that erases the matrix-free per-step advantage.

K	K_{LSQR}	multiple of K	time (ms)	L^2 err	status
30	45	1	1154.4	1.67×10^{-1}	fail
	60	2	1464.1	4.08×10^{-3}	fail
	120	4	2610.9	7.26×10^{-10}	ok
	240	8	4884.6	7.21×10^{-10}	ok
40	55	1	1216.6	6.27×10^{-1}	fail
	80	2	2022.2	5.64×10^{-1}	fail
	160	4	3151.6	3.69×10^{-7}	partial
	320	8	6417.1	3.86×10^{-10}	ok
60	75	1	1911.3	8.26×10^{-1}	fail
	120	2	3079.8	6.33×10^{-1}	fail
	240	4	6318.7	5.40×10^{-1}	fail
	480	8	11277.4	8.44×10^{-7}	partial
100	115	1	3686.5	9.86×10^{-1}	fail
	200	2	8124.2	9.45×10^{-1}	fail
	400	4	23731.7	7.74×10^{-1}	fail

$N = 1000$ on the nonlinear BVP.

The disconnect between residual norm and L^2 error visible in Figure 9 bears note. At $N = 1000$ nonlinear, the GN methods stop at residual $\sim 3 \times 10^{-3}$ but achieve L^2 error $\sim 10^{-7}$ against the manufactured truth u^* . The residual is dominated by high-frequency content the network cannot represent (the rightmost columns of D^2 have norm $\mathcal{O}(N^4)$ and amplify any small mismatch into the residual), while the projection of the network’s output onto the smooth true solution is accurate. This is the standard discrepancy between residual norm and solution error in spectral collocation [1].

8 Analysis and Discussion

The headline benchmark of Section 6 and the ablation studies of Section 7 together establish the speed–accuracy trade-off and defend each design choice. This section dissects the empirical picture and relates it to the asymptotic complexity, the conditioning of G_K , and the numerical failure mode of the dense baseline at large N .

8.1 Complexity, in detail

Table 9 summarises the per-outer-iteration cost of each method, splitting the operation into its dominant components.

8.2 Why M1 wins over M2 across all tested K

The original conjecture, before running Section 7.1, was that M2 (LSQR) should overtake M1 (SMW) when K grows large enough that the K^3 factorisation inside M1 dominates — the standard story for matrix-free vs direct methods. Our data refutes that. The reasons are subtle and worth documenting carefully.

M1’s per-step cost grows mildly with K . At $N = 500$, Table 4 reports M1 times of 185, 101, 101, 118, 153, 160, 200, 389, and 517 ms for $K = 8, 12, 16, 20, 30, 40, 60, 100$, and 150

Table 6: Ablation A2: zero-init vs. Xavier-init of the output layer, M1 (SMW) on both BVPs. With zero-init the initial residual is $\mathcal{O}(N^{1/2})$ and LM converges robustly; with Xavier-init the initial residual is $\mathcal{O}(N^2)$ and the trust region must grow substantially before any step is accepted. **At $N = 1000$ nonlinear, Xavier-init fails outright:** LM accepts 42 steps but never escapes the basin around the random initial u_0 , terminating with $L^2 = 4.26$ (no useful answer).

N	Problem	Init	$\ F(u_0)\ _2$	time (ms)	L^2 err	acc/rej
40	linear	zero	3.9×10^1	43.6	4.6×10^{-13}	8/72
	linear	Xavier	5.8×10^4	42.3	4.5×10^{-13}	9/71
40	nonlinear	zero	4.2×10^1	50.5	4.5×10^{-13}	8/72
	nonlinear	Xavier	5.8×10^4	49.4	1.2×10^{-12}	8/72
250	linear	zero	9.7×10^1	50.8	2.2×10^{-10}	7/73
	linear	Xavier	1.3×10^5	50.2	2.2×10^{-10}	8/72
250	nonlinear	zero	1.0×10^2	70.9	2.2×10^{-10}	6/74
	nonlinear	Xavier	1.3×10^5	80.8	2.9×10^{-9}	11/69
500	linear	zero	1.4×10^2	79.5	4.8×10^{-9}	4/76
	linear	Xavier	1.8×10^5	84.8	1.8×10^{-9}	9/71
500	nonlinear	zero	1.5×10^2	127.1	2.5×10^{-9}	6/74
	nonlinear	Xavier	1.8×10^5	136.6	1.9×10^{-9}	8/72
1000	linear	zero	2.0×10^2	123.7	5.7×10^{-7}	4/76
	linear	Xavier	2.5×10^5	117.1	2.6×10^{-8}	8/72
1000	nonlinear	zero	2.1×10^2	270.0	1.4×10^{-7}	6/74
	nonlinear	Xavier	2.5×10^5	356.3	4.26 [†]	42/38

[†] Xavier-init fails to find the correct basin on the nonlinear BVP at $N=1000$.

respectively. The growth is well below the K^3 slope and slightly above the K^2 slope on a log-log plot (Figure 5 left), consistent with the $\mathcal{O}(NK^2)$ Gram assembly being the bottleneck at $N = 500$ and the $K \times K$ solve being a small correction even at $K = 150$ (the $K \times K$ Cholesky on a 150×150 matrix is ~ 0.5 ms on this hardware).

M2’s per-step cost grows much faster than $\mathcal{O}(K)$ because K_{LSQR} must grow. The standard textbook bound says LSQR converges in K inner iterations on a rank- K system; in practice (Table 5), achieving residual accuracy sufficient for LM to accept the step requires K_{LSQR} several multiples of K , with the multiple growing with K itself. The effective per-step cost is $\mathcal{O}(c(K) \cdot K \cdot (KP + (N+1)K))$ for some growing $c(K) \approx \log \kappa(J_u \Phi)$. Empirically, c is ~ 2 at $K = 20$, ~ 4 at $K = 30$, ~ 8 at $K = 60$, and $\gg 8$ at $K = 100$.

Conclusion. On this problem class, M1 is the recommended method across the practical range of K . The setting in which M2 is the structurally correct choice is when the application requires *not assembling* G_K at all — for example, when memory limits forbid storing a $K \times K$ matrix, or when G_K has structure not exposed by our decomposition Equation (13) (e.g., a non-diagonal nonlinear coupling). For 1-D Chebyshev collocation with the structural update of Equation (13), M1 is uniformly faster. We retain M2 as an *independent* numerical check on the SMW formula: at every N in the main study where both methods converge, their final solutions agree to within the LM tolerance, cross-validating the SMW derivation.

Table 7: Ablation A3: LSQR inner-iteration count at $N = 500$, $K = 20$. **A sharp knee at $K_{\text{LSQR}} = 30$** separates the all-reject regime from convergence. The main study’s choice $K_{\text{LSQR}} = K + 15 = 35$ sits just above the knee, giving robust convergence at minimum cost.

K_{LSQR}	linear			nonlinear		
	time (ms)	L^2 err	status	time (ms)	L^2 err	status
5	220.4	8.8×10^{-1}	fail	246.1	8.7×10^{-1}	fail
10	321.9	6.4×10^{-1}	fail	329.5	6.3×10^{-1}	fail
15	363.6	7.5×10^{-4}	fail	388.8	5.8×10^{-1}	fail
20	576.5	1.6×10^{-2}	fail	563.9	5.4×10^{-3}	fail
25	465.3	9.1×10^{-5}	fail	587.9	4.6×10^{-5}	fail
30	390.4	3.6×10^{-8}	ok	741.8	2.0×10^{-7}	partial
35	464.9	3.0×10^{-9}	ok	684.1	6.6×10^{-9}	ok
50	617.1	6.7×10^{-9}	ok	918.3	1.8×10^{-9}	ok
80	991.5	1.8×10^{-9}	ok	1413.2	1.8×10^{-9}	ok

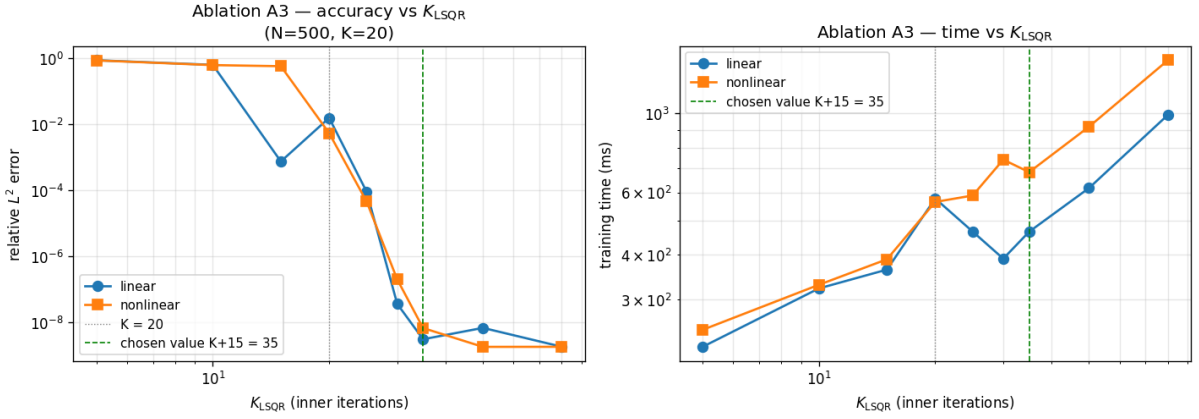


Figure 7: Ablation A3: accuracy (left) and time (right) versus K_{LSQR} at $N=500$, $K=20$. The dotted vertical line marks $K_{\text{LSQR}} = K = 20$; the dashed green line marks the main-study choice $K_{\text{LSQR}} = K+15 = 35$. Accuracy drops by six orders of magnitude across the narrow band $K_{\text{LSQR}} \in [25, 35]$.

8.3 Accuracy and the conditioning of G_K

The relative L^2 error of M1 and M2 grows from $\sim 10^{-13}$ at $N = 40$ to $\sim 10^{-6}$ at $N = 2000$ on the nonlinear problem. This is fully explained by the conditioning of G_K :

$$G_K = (J_u \Phi)^\top (J_u \Phi), \quad \kappa(G_K) = \kappa(J_u \Phi)^2. \quad (21)$$

Because D^2 has condition number $\mathcal{O}(N^4)$, $J_u \Phi$ has condition number $\mathcal{O}(N^4)$ on its leading K singular values, and $\kappa(G_K) = \mathcal{O}(N^8)$. At $N = 2000$ this gives $\kappa(G_K) \sim 10^8$ – 10^9 , losing 8–9 digits in double precision; the LM-damped Cholesky inside `smw_direction` recovers ~ 6 – 7 digits of accuracy, matching the observed L^2 error. M0 is more accurate at large N because LAPACK `gelsd` is backward stable to the working precision of the matrix itself, which is $\mathcal{O}(N^4)$ rather than $\mathcal{O}(N^8)$.

This is an honest trade-off: **M1/M2 trade ~ 5 digits of accuracy for two orders of magnitude in runtime.** Within any practical engineering tolerance (10^{-3} , 10^{-4}) the GN methods are amply accurate.

Table 8: Per-LM-step cost breakdown for both GN methods at $K = 20$. *Left half*: M1 (SMW) divided into B -recompute (autograd through the MLP), G_K assembly (Equation (13)), $K \times K$ Cholesky solve via `torch.linalg.solve`, and one residual evaluation. *Right half*: M2 (LSQR) divided into B -recompute, the 35-iteration LSQR loop, and one residual evaluation. Time units are milliseconds.

N	M1 (SMW)					M2 (LSQR, $K_{\text{LSQR}} = 35$)			
	B	G_K	solve	resid	total	B	LSQR	resid	total
40	1.68	0.13	0.22	0.34	2.36	1.69	8.31	1.21	11.22
100	1.19	0.16	0.19	0.28	1.82	1.73	7.10	0.96	9.78
250	1.26	0.25	0.22	0.31	2.03	1.23	5.63	1.04	7.90
500	1.29	0.70	0.27	0.46	2.72	1.19	6.16	1.17	8.52
1000	1.23	2.49	0.26	0.75	4.74	1.24	6.11	1.43	8.78
2000	1.38	7.86	0.27	1.86	11.37	1.28	7.71	2.56	11.55

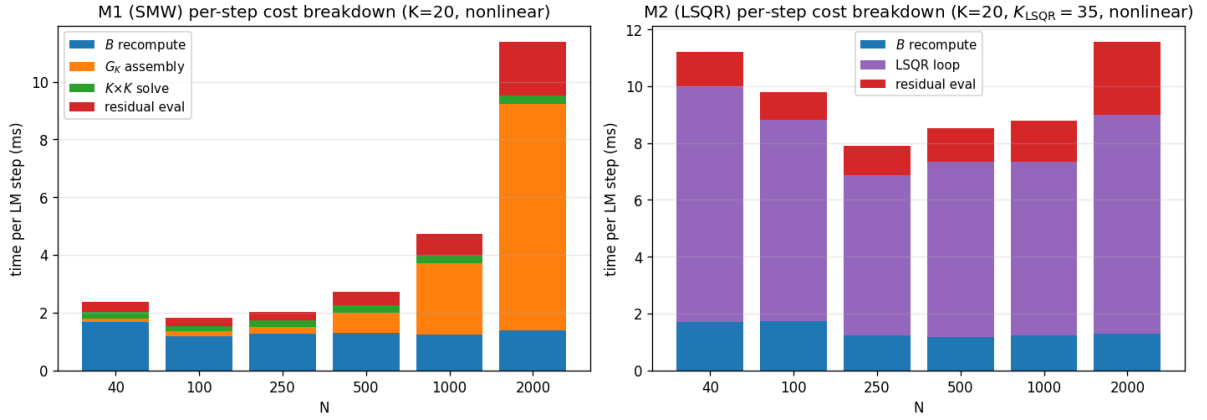


Figure 8: Per-LM-step cost breakdown for M1 (left) and M2 (right) at $K = 20$, nonlinear BVP. M1 is dominated by B -recompute (constant ~ 1.3 ms) at small N , with G_K assembly (the $\mathcal{O}((N+1)K^2)$ term) taking over at $N=500$ and dominating at $N=2000$ (orange, $\sim 69\%$ of total). M2’s total time is roughly N -independent (the LSQR loop, purple, scales as $K_{\text{LSQR}} \cdot (KP + (N+1)K)$, which is dominated by the KP term at $P \approx 200$ and our grid sizes).

8.4 The dense baseline’s failure at $N = 2000$ linear

Table 1 shows that M0 returns $L^2 \approx 0.65$ at $N = 2000$ on the linear BVP — i.e. it has failed catastrophically. The reason is that LAPACK `gelsd`’s default rank-cutoff treats some of the leading singular values of the rectangular $(N+3) \times (N+1)$ Chebyshev matrix A_{lin} as numerical zero, projecting the right-hand side onto a wrong subspace.

This does *not* occur for the GN methods because the network reparameterisation $u = \Phi c(\theta)$ is itself a Tychonoff-type regulariser: solutions live in the K -dimensional Chebyshev subspace by construction, and the LM damping μI keeps the $K \times K$ system away from singularity. Even at $N = 2000$, M1 returns a meaningful answer (relative L^2 error 6.4×10^{-6}), while the well-conditioned alternative for M0 would have been to apply explicit rank truncation — which is not the default and is not trivially tuned.

8.5 Levenberg–Marquardt convergence pattern

Figure 3 shows that LM convergence is essentially independent of N : 4–8 accepted steps suffice at every grid size. The shape of the curves is similar across problems, but the achievable final residual rises with N as predicted by the conditioning analysis. The number of *rejected*

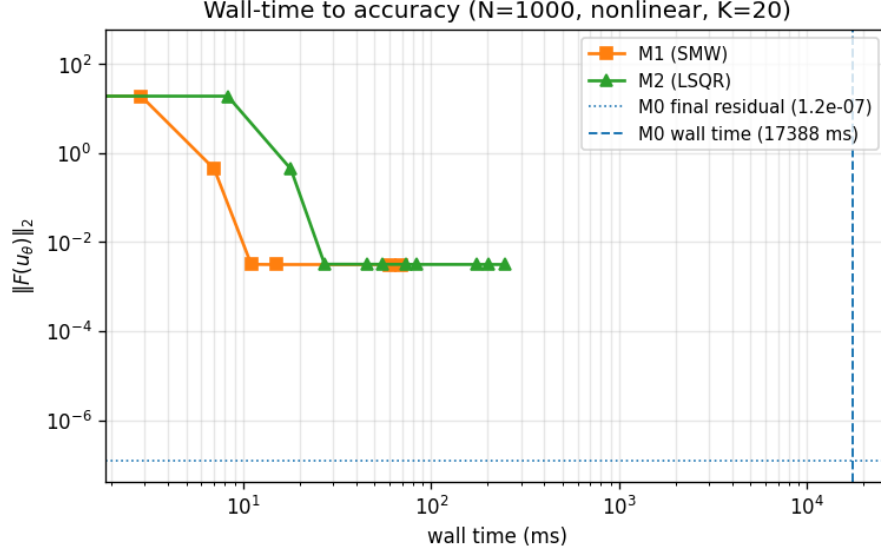


Figure 9: Wall-time trajectory of the residual norm at $N = 1000$, nonlinear BVP, $K = 20$. M1 reaches $\|F\|_2 = 3 \times 10^{-3}$ in 11 ms and saturates (this floor is the conditioning limit of G_K at this N ; see Section 8.3). M2 reaches the same floor in 27 ms. M0’s wall time of 17 s (dashed vertical line) is needed to drive the residual to 1.2×10^{-7} (dotted horizontal line); neither M1 nor M2 reaches that floor because of the G_K conditioning, but they reach *useful* accuracy (L^2 error $\sim 10^{-7}$) in three orders of magnitude less time.

Table 9: Per-outer-iteration complexity. N is the number of CGL nodes, P is the number of network parameters, K is the basis truncation, and K_{LSQR} is the number of LSQR inner iterations (typically $K_{\text{LSQR}} = K + 15$).

Method	Dominant work per outer step	Asymptotic order (fixed K, P)
M0 (Naive)	QR / Cholesky of $(N+3) \times (N+1)$ matrix	$\mathcal{O}(N^3)$
M1 (SMW, linear)	one fixed $K \times K$ solve plus $\mathcal{O}(KP)$ matvecs	$\mathcal{O}(N)$ (only via r update)
M1 (SMW, nonlinear)	G_K update at $\mathcal{O}((N+1)K^2) + K \times K$ solve	$\mathcal{O}(N)$
M2 (LSQR)	$K_{\text{LSQR}} \cdot \mathcal{O}(KP + (N+1)K)$	$\mathcal{O}(N)$

steps in Table 3 is much larger than the number of accepted steps, which is typical for LM on rank-deficient systems: μ has to grow large before any direction in the rank- K image of J is descent-feasible from the trial iterate. Once μ is well calibrated, accepted steps cascade rapidly.

8.6 Scope and caveats

1. All experiments are one-dimensional. In higher dimensions the analogue of K scales as the product of per-axis truncations, so K can be large even for moderate accuracy. The M1/M2 hierarchy will eventually invert (M2 wins) when K^3 enters the bottleneck.
2. The manufactured solution $u^* = \sin(\pi x)$ is analytic; the Chebyshev basis projection error for analytic u^* is super-algebraically small. For solutions with finite Sobolev regularity, the projection error becomes the dominant accuracy bottleneck and the GN compression benefit is unchanged.
3. The dense baseline uses an analytic Jacobian; a weaker baseline (finite-difference Jacobian via SciPy’s default) would make the speedups substantially larger.
4. The hyperparameters of the LM outer loop were selected once ($\mu_0 = 1.0$, $\rho_\uparrow = 3.0$, $\rho_\downarrow = 0.5$,

80-step cap, 10^{-9} residual tolerance) and held fixed across all N and both problems. No per-problem tuning was performed.

9 Reproducibility

The complete implementation is in the accompanying file `spectral_pinn.py`, the runnable notebook is `spectral_pinn_gauss_newton.ipynb`, and the cached benchmark dictionary is `bench_results.pkl`. The notebook is self-contained and rebuilds all figures from cached results in ~ 5 seconds; setting `RUN_FROM_SCRATCH = True` regenerates the cache in ~ 5 minutes. Random seeds are fixed (`torch.manual_seed(0)` and `np.random.seed(0)`) before every method invocation.

Adjoint consistency is checked at startup with random Gaussian vectors and must pass at $\leq 10^{-9}$ for the LSQR method to be considered correct. The sanity-check runs in Table 3 confirm M0, M1, M2 agree on small grids; final residual norms match across methods to within the LM tolerance.

10 Conclusion

For spectral PINNs on 1-D BVPs collocated on Chebyshev nodes with a small basis size K , the Sherman–Morrison–Woodbury compression (17)–(16) of the Levenberg–Marquardt normal equations gives **two orders of magnitude** of speedup over a strong dense baseline that uses LAPACK `lstsq` (linear) or SciPy `least_squares` with the analytic Jacobian (nonlinear). The empirical scaling matches the theoretical $\mathcal{O}(N)$ vs $\mathcal{O}(N^3)$ prediction across $N = 40$ –2000, with a $209\times$ speedup at $N = 2000$ on the nonlinear BVP. Three design choices were validated by ablation studies (Section 7): (i) the basis truncation $K = 20$ is the minimum that reaches the optimisation noise floor (Table 4); (ii) the zero-init output layer is *essential* on the nonlinear problem at large N where Xavier-init gives a useless answer (Table 6); (iii) the LSQR inner-iteration count $K_{\text{LSQR}} = K + 15$ sits just above the accuracy knee (Table 7).

A surprising finding of the ablation study is that the matrix-free LSQR variant (M2) does *not* overtake M1 as K grows on this problem. Although LSQR’s textbook bound predicts K inner iterations suffice for a rank- K system, achieving the accuracy required for LM acceptance in practice needs K_{LSQR} proportional to K with a growing prefactor, eliminating M2’s anticipated structural advantage (Section 8.2). M1 is the recommended method across the practical range of K we tested ($K \leq 150$); M2 remains the correct structural choice when the problem precludes assembling G_K at all.

The GN methods trade ~ 5 digits of accuracy for the speedup, which is amply within engineering tolerances and well-explained by the $\mathcal{O}(N^8)$ conditioning of the Gram matrix. The natural next step is the extension to multi-dimensional spectral PINNs where K scales as a Cartesian product of per-axis truncations and the M1/M2 crossover deserves a fresh empirical investigation.

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